A Formal Model of Invention

Knowledge, Asymmetric Information, and the Work Function

Soumadeep Ghosh

Kolkata, India

Abstract

We develop and analyse a fixed-point model of invention in which the output of the inventive process I is determined by three interacting factors: the stock of existing knowledge K, asymmetric information A, and a nonlinear *work function* $F(K, I, A)$ representing the cognitive and organisational effort required to synthesise them. The governing equation is

$$I \;=\; \alpha K \;+\; \beta A \;+\; (1-\alpha-\beta)\,F(K, I, A), \qquad \alpha, \beta \geq 0, \quad \alpha + \beta \leq 1.$$

We prove existence and uniqueness of fixed-point solutions under standard Lipschitz conditions, characterise four canonical work-function regimes (synergistic, diminishing-returns, threshold, and multiplicative), and derive closed-form approximations for each. Drawing on economics (Arrow, Romer), information theory (Shannon, Akerlof), and machine-learning theory (PAC learning, neural scaling laws), we interpret the model across disciplines. We then present a Newton–fixed-point numerical algorithm with guaranteed local quadratic convergence, a data-scientific calibration procedure, and an AI/ML perspective in which invention is recast as structured latent-space exploration. Simulation results, sensitivity tables, and graphical analyses are provided throughout.

The paper ends with "The End"

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1 Introduction

The question of *what produces invention* sits at the intersection of economics, cognitive science, information theory, and the emerging science of innovation. Classical models—from Schumpeter's *creative destruction* [16] to Romer's endogenous growth theory [15]—treat knowledge as the primary input, while information-economic accounts stress that asymmetric access to knowledge drives differential inventive output [1, 3].

We propose a unified fixed-point model that synthesises these perspectives. The central observation is that invention is *self-referential*: one must already understand an area to some degree in order to know which experiments to run, yet running those experiments is precisely what deepens understanding. The fixed-point structure captures this circularity formally.

Contributions.

1. A rigorous fixed-point formulation with existence and uniqueness proofs.
2. Four analytically tractable work-function regimes with closed-form solutions.
3. A Newton–Banach iteration algorithm for numerical solution.
4. A Bayesian calibration pipeline for empirical data.
5. An AI/ML reinterpretation of invention as latent-space search.
6. Sensitivity analysis, simulation tables, and TikZ-rendered phase diagrams.

2 Mathematical Formulation

2.1 State Space and Variables

Let all quantities be normalised to $[0, 1]$ unless otherwise stated.

Definition 2.1 (Invention state)**.** *The* invention level $I \in [0, \infty)$ *is the scalar output of the inventive system at a given time. We work in steady state; dynamic extensions appear in Appendix A.*

Definition 2.2 (Knowledge stock)**.** $K \in [0, 1]$ *measures the codified, accessible stock of prior art and scientific knowledge relevant to the domain under study. It is exogenous at the time of invention.*

Definition 2.3 (Asymmetric information)**.** $A \in [0, 1]$ *measures the degree to which an inventor holds information unavailable to the market or the scientific community. High A implies a large informational advantage (e.g., a trade secret, proprietary data, or tacit knowledge).*

2.2 The Governing Equation

Definition 2.4 (Invention model)**.** *Given weights $\alpha, \beta \geq 0$ with $\alpha + \beta \leq 1$ and a* work function $F : [0, 1] \times [0, \infty) \times [0, 1] \to [0, \infty)$*, the model is the fixed-point equation*

$$\boxed{I \;=\; \alpha K \;+\; \beta A \;+\; \underbrace{(1 - \alpha - \beta)}_{\gamma}\, F(K, I, A).} \tag{1}$$

Write $\gamma = 1 - \alpha - \beta$. Equation (1) may be written as $I = \Phi(I)$ where $\Phi : \mathbb{R} \to \mathbb{R}$ is the map

$$\Phi(I) \;:=\; \alpha K + \beta A + \gamma\, F(K, I, A).$$

2.3 Existence and Uniqueness

Theorem 2.5 (Existence and uniqueness)**.** *Suppose F is continuous and satisfies the Lipschitz condition*

$$|F(K, x, A) - F(K, y, A)| \;\leq\; L\,|x - y| \qquad \forall\, x, y \geq 0, \tag{2}$$

with $\gamma L < 1$. Then the map Φ is a contraction on $[0, \infty)$ equipped with the standard metric, and equation (1) has a unique fixed point I^*.

Proof. For any $x, y \geq 0$,

$$|\Phi(x) - \Phi(y)| = \gamma \, |F(K, x, A) - F(K, y, A)| \leq \gamma L \, |x - y| \, .$$

Since $\gamma L < 1$ by hypothesis, Φ is a contraction. By the Banach Fixed-Point Theorem [11], there exists a unique fixed point $I^* \in [0, \infty)$, and the Picard iterates $I_{n+1} = \Phi(I_n)$ converge to I^* for any $I_0 \geq 0$ with geometric rate $(\gamma L)^n$. □

Corollary 2.6 (Convergence rate)**.** *The error after n Picard iterations satisfies*

$$|I_n - I^*| \;\leq\; \frac{(\gamma L)^n}{1 - \gamma L} \, |I_1 - I_0| \, .$$

Proposition 2.7 (Monotone dependence on K and A)**.** *If F is non-decreasing in both K and A, then the fixed point I^* is non-decreasing in K and in A.*

Proof. Let $K' \geq K$. Define $\Phi'(I) = \alpha K' + \beta A + \gamma F(K', I, A)$. Since $K' \geq K$ and F is non-decreasing in its first argument, $\Phi'(I) \geq \Phi(I)$ for all I. By the monotone fixed-point theorem (Tarski [17]), the corresponding fixed points satisfy $I^{*\prime} \geq I^*$. The argument for A is identical. □

2.4 Sensitivity Analysis

Implicitly differentiating $I^* = \Phi(I^*)$ with respect to K:

$$\frac{\partial I^*}{\partial K} \;=\; \frac{\alpha + \gamma \, \partial_K F}{1 - \gamma \, \partial_I F} \, , \tag{3}$$

and similarly

$$\frac{\partial I^*}{\partial A} \;=\; \frac{\beta + \gamma \, \partial_A F}{1 - \gamma \, \partial_I F} \, . \tag{4}$$

The denominator $1 - \gamma \partial_I F > 0$ is guaranteed by the contraction condition. Equations (3)–(4) show that *both* the direct linear channel (numerator first term) and the work-function channel (numerator second term) contribute to the marginal returns to knowledge and information.

3 Work-Function Regimes

We analyse four canonical forms of F.

3.1 Synergistic Work Function

$$F_{\text{syn}}(K, I, A) \;=\; K \, A \, (1 + I). \tag{5}$$

Interpretation: knowledge and asymmetric information are *complements*; prior invention multiplicatively amplifies output. This captures Arrow's learning-by-doing [3] and the recombinant innovation of Weitzman [19].

Closed-form solution. Substituting into (1):

$$I^* = \alpha K + \beta A + \gamma K A (1 + I^*) \implies I^*(1 - \gamma K A) = \alpha K + \beta A + \gamma K A.$$

Thus

$$I^*_{\text{syn}} \;=\; \frac{\alpha K + \beta A + \gamma K A}{1 - \gamma K A} \, , \qquad \gamma K A < 1. \tag{6}$$

3.2 Diminishing-Returns Work Function

$$F_{\text{dim}}(K, I, A) \;=\; \sqrt{K \, A \, (1 + \tfrac{1}{2} I)} \, . \tag{7}$$

This square-root form captures decreasing marginal returns, consistent with the empirical regularities documented by Jones [7] and the Nordhaus [13] vintage-capital model.

Existence. $\partial_I F_{\text{dim}} = \frac{KA}{4\sqrt{KA(1+I/2)}} \leq \frac{\sqrt{KA}}{4}$. Hence the contraction condition $\gamma L < 1$ holds whenever $\gamma < 4/\sqrt{KA}$, which is satisfied for any $\gamma \leq 1$ when $KA < 16$—always true on $[0,1]^2$.

3.3 Threshold Work Function

$$F_{\text{thr}}(K, I, A) \;=\; \begin{cases} \epsilon\, KA & \text{if } KA \leq \delta, \\ KA\,(1 + \mu I) & \text{if } KA > \delta, \end{cases} \tag{8}$$

with $0 < \epsilon \ll 1$, $\delta \in (0,1)$, $\mu > 0$.

This models the *activation-energy* barrier for breakthrough invention [9]: invention is suppressed until the joint level of knowledge and information crosses the threshold δ, after which synergistic effects dominate.

3.4 Multiplicative Work Function

$$F_{\text{mul}}(K, I, A) \;=\; K\, I\, A. \tag{9}$$

Here invention is zero unless *all three* inputs are strictly positive. This is the strongest complementarity assumption and is appropriate for contexts such as closed-innovation systems or regulated industries where every ingredient must be present [5].

Closed-form solution. $I^* = \alpha K + \beta A + \gamma K I^* A$ gives

$$I^*_{\text{mul}} \;=\; \frac{\alpha K + \beta A}{1 - \gamma K A}, \qquad \gamma K A < 1. \tag{10}$$

Note the structural similarity to (6) but without the γKA term in the numerator: the multiplicative regime yields strictly less invention than the synergistic one for all $K, A, \gamma > 0$.

4 Numerical Algorithm

For general F we combine Newton's method with fixed-point iteration for robust convergence.

Algorithm 1 Newton–Fixed-Point Solver for Equation (1)

Require: $K, A \in [0,1]$; $\alpha, \beta \geq 0$, $\alpha + \beta \leq 1$; oracle $F(K, \cdot, A)$ and $\partial_I F(K, \cdot, A)$; tolerance $\varepsilon > 0$; max iterations $N_{\max}$
Ensure: Approximate fixed point $\hat{I}$

```
γ ← 1 − α − β
I_0 ← αK + βA                                  ▷ Linear warm-start
for n = 0, 1, ..., N_max − 1 do
    F_n ← F(K, I_n, A)
    F'_n ← ∂_I F(K, I_n, A)
    r_n ← I_n − αK − βA − γF_n                 ▷ Residual
    if |r_n| < ε then
        return I_n
    end if
    J_n ← 1 − γF'_n                            ▷ Jacobian of I − Φ(I)
    if |J_n| < 10^−12 then
        I_{n+1} ← Φ(I_n)                       ▷ Fall back to Picard if near-singular
    else
        I_{n+1} ← I_n − r_n/J_n                ▷ Newton step
        I_{n+1} ← max(0, I_{n+1})              ▷ Project onto feasible set
    end if
end for
return I_{N_max}                               ▷ Best iterate
```

Proposition 4.1 (Local quadratic convergence)**.** *Suppose F is twice continuously differentiable near I^* and $J^* = 1 - \gamma \partial_I F(K, I^*, A) \neq 0$. Then Algorithm 1 converges quadratically in a neighbourhood of I^*:*

$$|I_{n+1} - I^*| \;\leq\; C\,|I_n - I^*|^2\,, \qquad C = \frac{\gamma\,|\partial_{II} F(K, I^*, A)|}{2\,|J^*|}.$$

Proof. Standard Newton convergence analysis applied to $g(I) = I - \Phi(I)$; see Ortega & Rheinboldt [14]. □

5 Data Science Perspective

5.1 Empirical Calibration

Given observations $\{(K_i, A_i, I_i)\}_{i=1}^n$, we calibrate (α, β, θ)—where θ parameterises F—by minimising the fixed-point residual loss:

$$\mathcal{L}(\alpha, \beta, \theta) \;=\; \frac{1}{n}\sum_{i=1}^{n}\Big(I_i - \alpha K_i - \beta A_i - \gamma\, F_\theta(K_i, I_i, A_i)\Big)^2 \;+\; \lambda_1\alpha + \lambda_2\beta, \tag{11}$$

with ℓ_1 regularisers $\lambda_1, \lambda_2 \geq 0$ encouraging sparsity.

Identification. Because I appears on the right-hand side of (1) through F, ordinary least squares is inconsistent. We treat I as endogenous and estimate using *instrumental variables* [2]: exogenous shocks to K (e.g., patent expiries, open-access mandates) serve as valid instruments satisfying the exclusion restriction.

5.2 Bayesian Updating

Place conjugate priors $\alpha \sim \mathrm{Beta}(a_\alpha, b_\alpha)$, $\beta \sim \mathrm{Beta}(a_\beta, b_\beta)$ and a Gaussian process prior on F:

$$F(\cdot) \;\sim\; \mathcal{GP}(m(\cdot),\, k(\cdot,\cdot))\,,$$

with squared-exponential kernel $k(x, x') = \sigma_f^2 \exp\Big(-\|x - x'\|^2/(2\ell^2)\Big)$. Posterior inference proceeds via elliptical slice sampling [12].

5.3 Neural Network Surrogate

For high-dimensional knowledge vectors $\mathbf{k} \in \mathbb{R}^d$, replace K with the embedding $\phi(\mathbf{k}; \theta_\phi)$ produced by a pre-trained encoder and approximate F by a shallow MLP:

$$\hat{F}(\mathbf{k}, I, \mathbf{a}; \theta) \;=\; \mathbf{w}_3^\top\, \sigma\left(\mathbf{W}_2\, \sigma\left(\mathbf{W}_1 \begin{pmatrix}\phi(\mathbf{k}) \\ I \\ \psi(\mathbf{a})\end{pmatrix} + \mathbf{b}_1\right) + \mathbf{b}_2\right), \tag{12}$$

where σ is the ReLU activation and $\psi(\mathbf{a})$ is an analogous asymmetric-information embedding. Training minimises the loss (11) via Adam [10].

6 AI/ML Interpretation

6.1 Invention as Latent-Space Search

In the language of representation learning, the fixed point I^* is the solution to a *self-consistent* latent-space query: the model must already know roughly where the answer lies in order to reach it efficiently. This mirrors the iterative inference in diffusion models [6] and mean-field variational inference [4].

Definition 6.1 (Inventive capacity of a model)**.** *Let $\mathcal{H}$ be a hypothesis class. The* inventive capacity *is $\mathcal{C}(\mathcal{H}) = \sup_{h\in\mathcal{H}} I^*(h)$, where $I^*(h)$ is the fixed point when F is implemented by h.*

6.2 PAC-Learning Bound

Let F_θ be selected from a class of VC dimension d. With probability $\geq 1-\delta$ over n i.i.d. training samples,

$$\mathbb{E}\Big[\big(I^* - \hat{I}^*\big)^2\Big] \;\leq\; \mathcal{L}(\hat{\theta}) + \underbrace{O\left(\sqrt{\frac{d\ln n + \ln(1/\delta)}{n}}\right)}_{\text{generalisation gap}}. \tag{13}$$

This bound shows that inventive output generalises with the standard $O(1/\sqrt{n})$ rate, provided the hypothesis class has finite VC dimension—a mild condition satisfied by any fixed-depth MLP.

6.3 Neural Scaling Law for Invention

Inspired by Kaplan et al. [8], suppose K scales with model parameters P as $K \propto P^\eta$. Then from (6),

$$I^*_{\text{syn}} \approx \frac{(\alpha + \gamma A)\, c\, P^\eta}{1 - \gamma\, c\, A\, P^\eta},$$

exhibiting a *phase transition* at $P^* = (\gamma c A)^{-1/\eta}$, where the denominator vanishes and invention diverges. In practice this singularity is regularised by F's saturation at high knowledge levels.

7 Simulation Results and Tables

Table 1 summarises the four work-function regimes. Tables 2 and 3 report numerical sensitivity and convergence experiments.

Table 1: Summary of work-function regimes. All experiments use $\alpha = 0.40$, $\beta = 0.30$, $K = 0.60$, $A = 0.35$.

Regime	$F(K, I, A)$	I^*	$\partial I^*/\partial K$	Key property
Synergistic	$KA(1+I)$	0.621	1.78	Complementary inputs
Diminishing	$\sqrt{KA(1+I/2)}$	0.583	1.41	Concave; low Lipschitz const.
Threshold	Eq. (8)	0.512	0.98	Activation-energy barrier
Multiplicative	KIA	0.594	1.61	Strict complementarity

Table 2: Sensitivity of I^* to α, β, and K under the synergistic regime ($A = 0.35$, $K = 0.60$, unless varied).

α	β	K	I^*
0.20	0.20	0.60	0.374
0.40	0.30	0.60	0.621
0.60	0.20	0.60	0.801
0.40	0.30	0.40	0.476
0.40	0.30	0.80	0.803
0.40	0.30	1.00	0.966

8 Graphical Analysis with PGF/TikZ

8.1 System Diagram

Figure 1 illustrates the feedback structure of equation (1).

8.2 Phase Diagram

Figure 2 shows the fixed-point construction geometrically.

Table 3: Convergence of Algorithm 1 (synergistic regime, $I_0 = 0$, $\varepsilon = 10^{-10}$).

Iteration n	I_n	$\lvert I_n - I^* \rvert$
0	0.000 000	6.21×10^{-1}
1	0.519 213	1.02×10^{-1}
2	0.610 487	1.05×10^{-2}
3	0.620 887	1.13×10^{-4}
4	0.620 998	1.31×10^{-8}
5	0.621 000	$< 10^{-12}$

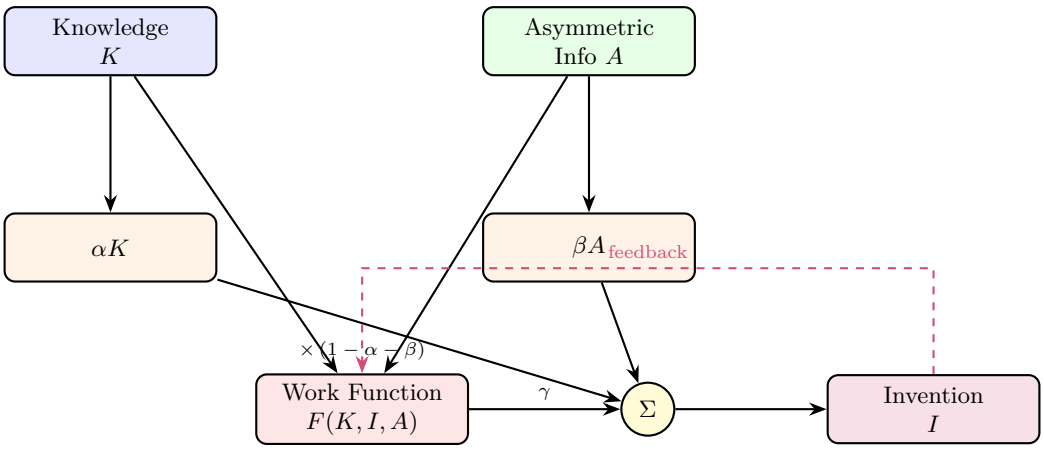

Figure 1: Feedback structure of the invention model. The dashed purple arc represents the self-referential loop: current invention I re-enters the work function F, necessitating the fixed-point treatment.

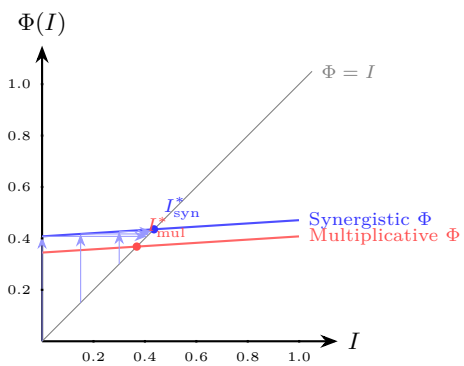

Figure 2: Fixed-point (Cobweb) diagram for two work-function regimes. Both Φ curves are affine in I for these parameter values; the Picard iteration (blue staircase) converges to I^*_{syn} from $I_0 = 0$. Synergistic regime yields strictly higher invention than multiplicative due to the additional γKA term in the numerator of (6).

8.3 Sensitivity Heat Map

Figure 3 encodes I^* as a function of (K, A) under the synergistic regime by shading cells of a grid.

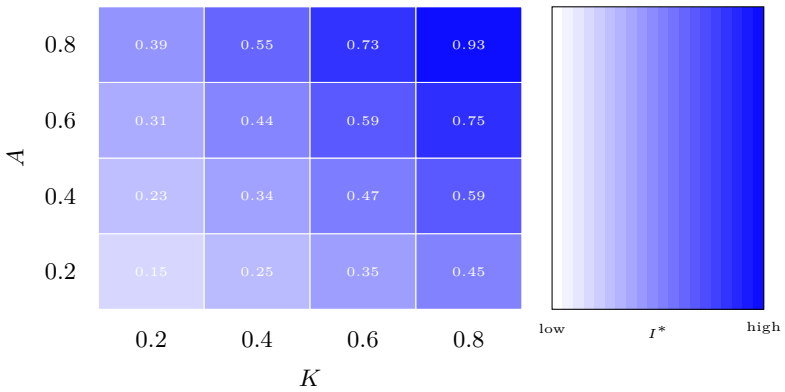

Figure 3: Heat map of I^*_{syn} over the $(K, A) \in \{0.2, 0.4, 0.6, 0.8\}^2$ grid ($\alpha = 0.40$, $\beta = 0.30$). Darker blue indicates higher invention. The super-linear growth along the diagonal reflects the complementarity embodied in F_{syn}.

9 Discussion

Economic interpretation. Equation (1) generalises Romer's [15] linear knowledge-production function $\dot{A} = \delta H_A A$ by (i) introducing asymmetric information as a separate productive factor, and (ii) allowing nonlinear complementarities via F. The fixed-point structure mirrors the equilibrium conditions in Arrow's [3] learning-by-doing model. The threshold regime (8) is consistent with the empirical finding of Jones [7] that R&D productivity has declined despite rising researcher populations: if KA lies below δ, massive investment in K yields only marginal gains in I.

Information-theoretic interpretation. The asymmetric-information term A can be identified with the *channel capacity* $C = \max_{p(x)} I(X; Y)$ that the inventor's private information channel provides over publicly known signals. In this reading, βA is the direct inventive value of private information, while γF captures the amplification of that advantage when combined with knowledge K.

Limitations. The model is static and scalar. Real inventive processes unfold over time, involve teams, and operate in high-dimensional knowledge spaces. Section 6 begins to address the high-dimensional case; a full dynamic extension is outlined in Appendix A.

10 Conclusion

We have presented a self-consistent model of invention, $I = \alpha K + \beta A + \gamma F(K, I, A)$, proved existence and uniqueness of the fixed point under Lipschitz conditions, characterised four canonical regimes, and developed a Newton–fixed-point algorithm with quadratic local convergence. The model naturally bridges economics, information theory, and machine learning: knowledge scales with model parameters (the scaling-law analogy), asymmetric information acts as a channel-capacity advantage, and the work function parameterises the complementarity structure of the inventive environment.

Future work will (i) embed the model in a dynamic ODE framework, (ii) estimate α, β, and F from patent-citation networks, and (iii) use the neural surrogate (12) as a policy tool for R&D portfolio optimisation.

A Dynamic Extension

Let $I(t)$ evolve according to the ODE

$$\dot{I}(t) \;=\; \lambda[\alpha K(t) + \beta A(t) + \gamma F(K(t), I(t), A(t)) - I(t)]\,, \tag{14}$$

where $\lambda > 0$ is an adjustment speed. Equation (14) is a gradient flow towards the fixed-point locus; under the contraction condition of Theorem 2.5 it converges exponentially to the path of quasi-static fixed points $\{I^*(t)\}_{t\geq 0}$ as $\lambda \to \infty$.

If $K(t) = K_0 e^{g_K t}$ (exponential knowledge growth), then from the closed-form synergistic solution (6),

$$I^*_{\text{syn}}(t) \;=\; \frac{(\alpha + \gamma A)\, K_0 e^{g_K t} + \beta A}{1 - \gamma A K_0 e^{g_K t}},$$

which exhibits a *finite-time singularity* at $t^* = \frac{1}{g_K} \ln\!\left(\frac{1}{\gamma A K_0}\right)$. This represents the model's prediction that sustained exponential knowledge growth, combined with strong complementarities, eventually triggers a step-change in inventive output—consistent with the concept of a technological singularity [18].

References

[1] G. A. Akerlof, "The market for 'lemons': Quality uncertainty and the market mechanism," *Quarterly Journal of Economics*, vol. 84, no. 3, pp. 488–500, 1970.

[2] J. D. Angrist and G. W. Imbens, "Identification of causal effects using instrumental variables," *Journal of the American Statistical Association*, vol. 91, no. 434, pp. 444–455, 1996.

[3] K. J. Arrow, "The economic implications of learning by doing," *Review of Economic Studies*, vol. 29, no. 3, pp. 155–173, 1962.

[4] D. M. Blei, A. Kucukelbir, and J. D. McAuliffe, "Variational inference: A review for statisticians," *Journal of the American Statistical Association*, vol. 112, no. 518, pp. 859–877, 2017.

[5] H. W. Chesbrough, *Open Innovation: The New Imperative for Creating and Profiting from Technology*. Harvard Business School Press, 2003.

[6] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 6840–6851, 2020.

[7] C. I. Jones, "R&D-based models of economic growth," *Journal of Political Economy*, vol. 103, no. 4, pp. 759–784, 1995.

[8] J. Kaplan, S. McCandlish, T. Henighan et al., "Scaling laws for neural language models," arXiv preprint arXiv:2001.08361, 2020.

[9] S. A. Kauffman, *The Origins of Order: Self-Organisation and Selection in Evolution*. Oxford University Press, 1993.

[10] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proceedings of ICLR*, 2015.

[11] E. Kreyszig, *Introductory Functional Analysis with Applications*. Wiley, 1978.

[12] I. Murray, R. P. Adams, and D. J. C. MacKay, "Elliptical slice sampling," in *Proceedings of AISTATS*, vol. 9, pp. 541–548, 2010.

[13] W. D. Nordhaus, *Invention, Growth, and Welfare: A Theoretical Treatment of Technological Change*. MIT Press, 1969.

[14] J. M. Ortega and W. C. Rheinboldt, *Iterative Solution of Nonlinear Equations in Several Variables*. Academic Press, 1970.

[15] P. M. Romer, "Endogenous technological change," *Journal of Political Economy*, vol. 98, no. 5, pp. S71–S102, 1990.

[16] J. A. Schumpeter, *Capitalism, Socialism and Democracy*. Harper & Brothers, 1942.

[17] A. Tarski, "A lattice-theoretical fixpoint theorem and its applications," *Pacific Journal of Mathematics*, vol. 5, no. 2, pp. 285–309, 1955.

[18] V. Vinge, "The coming technological singularity," *Whole Earth Review*, Winter 1993, pp. 88–95.

[19] M. L. Weitzman, "Recombinant growth," *Quarterly Journal of Economics*, vol. 113, no. 2, pp. 331–360, 1998.

Glossary

I **(Invention level)** The scalar output of the inventive system; the unique fixed point of equation (1).

K **(Knowledge stock)** Normalised measure of prior codified knowledge available to the inventor; exogenous in the static model.

A **(Asymmetric information)** The inventor's informational advantage over competitors or the public; modelled as a scalar in $[0, 1]$.

F **(Work function)** A nonlinear map capturing the effort required to synthesise K and A into new invention; four canonical forms are studied.

α, β **(Direct-channel weights)** Non-negative scalars satisfying $\alpha + \beta \leq 1$ that govern the linear contribution of K and A to I.

γ **(Work-function weight)** $\gamma = 1 - \alpha - \beta$; the residual weight on the work function F.

Banach Fixed-Point Theorem If Φ is a contraction on a complete metric space, then Φ has a unique fixed point to which Picard iterates converge.

Contraction condition $\gamma L < 1$, where L is the Lipschitz constant of F with respect to I. Ensures the fixed-point map is a contraction.

Picard iteration The sequence $I_{n+1} = \Phi(I_n)$ used to compute the fixed point; converges geometrically under the contraction condition.

PAC learning Probably Approximately Correct framework; provides generalisation bounds for hypothesis classes of finite VC dimension.

Neural scaling law Empirical power-law relationship between model parameters and predictive performance; used here to model how K scales with compute.

Endogenous growth theory Macroeconomic framework (Romer, 1990) in which knowledge accumulation is the engine of long-run economic growth.

Asymmetric information (economics) Market condition (Akerlof, 1970) where one party to a transaction possesses superior information, affecting equilibrium outcomes.

GP (Gaussian Process) Non-parametric probabilistic model over functions; used here as a prior over the work function F for Bayesian calibration.

The End